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Autorzy / authors	Konopielko Łukasz, Wytrębowicz Jacek
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Staking – Technological and Economic Challenges Deposit-as-a-Service as an Alternative to Zero Interest Rates

Łukasz KONOPIELKO

Łazarski University, Warsaw, POLAND, e-mail: lukasz.konopielko@lazarski.pl
ORCID: 0000-0001-5557-0965

Jacek WYTRĘBOWICZ

Warsaw University of Technology, POLAND, e-mail: J.Wytrebowicz@ii.pw.edu.pl
ORCID: 0000-0003-3804-8133

Abstract

The rapid development of blockchain technologies over recent years has formed a basis for numerous services that emulate and facilitate already existing functions of financial intermediaries, including money transfers, payments and currency exchange. This paper analyses development of staking, being a derivative of technical developments with cryptocurrencies. The background for the emergence of staking-oriented services is provided, both from its internal mechanism as well as user/investor side. Similarity to traditional deposit function is discussed and evaluated in the light of alternative investment opportunities on unregulated markets. Although being technically sophisticated, staking provides an interesting opportunity for risk-loving investors.

Keywords: Staking, blockchain, investment, cryptocurrencies.

Introduction

The staking is a deposit service offered for cryptocurrency holders. It may provide relatively high interest comparing to traditional banking deposits based on fiat money. Staking has emerged due to the rising popularity of blockchain applications and the evolution of their technical basis. This paper aims to give some insights into the technical basis for staking and analyse the economic perspective as an alternative investment instrument. Understanding technical basis and the knowledge about its economic state are crucial for any reasoning and predictions that whoever would undertake. The paper concludes with the short comparative reviews of staking vis-a-vis “old fashioned” deposit services.

Technical Basis

Currently, there are a lot of working blockchains supporting cryptocurrencies. A cryptocurrency is represented as tokens registered on a blockchain. However, not every such token serves as cryptocurrency. The tokens can be used to stand for different kind of assets or rights. The blockchain network consists of computing nodes that store a transaction ledger in a distributed, decentralised and safe way. A distributed storage eliminates the risk of data loss due to a failure and provides high availability. Decentralised storage eliminates the risk of corruption or other misbehaviour of the party that owns and maintains the storage. Safety means that any registered data cannot be altered, and only cryptographically authenticated users can be the participants of a transaction.

The transactions are grouped into blocks. Every block header contains a root of Merkle tree of all transactions belonging to the block and the hash of the previous block header, see Fig 1. A hash is a cryptographic function that maps arbitrary size data to fixed-size value (SHA2, SHA3, BLAKE3 are example hash functions). In consequence, the transactions registered on the blockchain are immutable. Any attempt to modify a registered transaction would have to recalculate some hashes in the Merkle tree, the hash of that block header, and all hashes in the subsequent headers. Moreover, such a change should be done in all nodes belonging to the blockchain network. There are several economic and technical means to protect the blockchain against such modification. A consensus mechanism is implemented in order to solve the so-called double-spending problem. New transactions are verified and added to a pool. Next, a new block is created from transactions taken from the pool. The nodes have to agree on the new block while adding it to the chain of blocks. An incorrect or conflicting transaction cannot be included in the blockchain. The consensus mechanism allows the blockchain network to validate and approve or disapprove transactions before storing them to make the blockchain faultless.

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Gathering transactions into a chain of blocks is not the only option. A distributed ledger can have the form of a directed acyclic graph (DAG) of transaction blocks, e.g. IOTA (www.iota.org) and NANO (nano.org) ledgers. However, blockchain structure is prevalent, and further, we will not distinguish DAGs and blockchains. A simple transaction defines a transfer of tokens, and a more complex transaction can define some conditions for such a transfer. The transaction is a programming script that can be considered as an immutable contract.

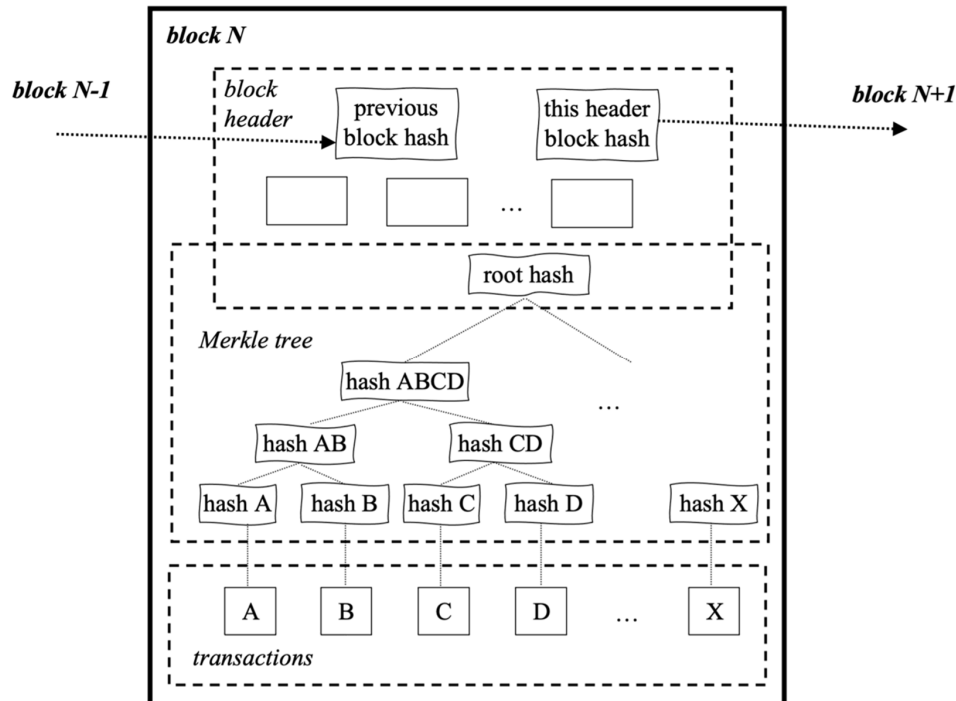


Fig 1. A simplified structure of a block

Nowadays, many blockchains provide support for smart contracts, i.e. user-defined immutable contracts registered on the network nodes together with the ledger. Any creation, execution or removal of a smart contract is registered as a transaction. Ethereum has been the first blockchain offering that feature. Smart contracts can be used to build a so-called decentralised application (DApp), where a decentralised peer-to-peer network (a blockchain network) is a virtual machine executing the backend code (smart contracts) of the application. The frontend code of DApp is a user interface that can be run on a web server or can be built into a crypto-wallet application. DApps can provide their own tokens and can be used in very different domains, binding their users by agreements written as smart contracts. DApps have found numerous adoptions in gambling and finance. The latter is called decentralised finance (DeFi). They allow people to lend or borrow funds, speculate on price movements on a range of assets, trade cryptocurrencies, insure against risks, and earn interest in savings-like accounts. The tokens created by a DeFi DApp can be a cryptocurrency that is accepted on exchanges. Some DeFi DApps utilise staked capital, based on their tokens, as “insurance funds”, for on-chain lending or margin trading. Harvey, Ramachandran and Santoro (2020) provide a comprehensive overview of the DeFi landscape.

Not only traditional enterprises but also decentralised autonomous organisations (DAOs) are involved in the development of DApps. DAO's business rules are defined by smart contracts, and its financial transaction record is registered on a blockchain. Examples of such DAOs are MakerDAO (makerdao.com) and Compound (compound.finance), which provide lending pools. They utilise a staking token that represents both on-chain transaction fees as well as a debt instrument. Another DAO, Synthetix provides margin trading services.

The consensus mechanisms, which are needed to guarantee the honest behaviour of nodes, are numerous. They evolve, and new ones are still being proposed. Xiao *et al.* (2020) and Ferdous *et al.* (2020) give good surveys of them. The first implemented consensus mechanism, applied in the bitcoin blockchain, uses the Proof-of-Work (PoW) principle. Cryptocurrency blockchains employ mainly PoW or Proof-of-Stake (PoS) principles. Most consensus mechanisms based on PoW or PoS allow a node to earn some cryptocurrencies as a prime for a newly added block to the blockchain. Each time it is a different node, and the probability of being the awarded one depends on the computation power of the node in the case of PoW or on the stake put on deposit by the node's owner in the case of PoS. Most cryptocurrency blockchains work so that an amount of cryptocurrency is created by the newly added block as the prime for the node's owner. For that reason, the block is called, in the blockchain jargon, a mined or a minted block for PoW and PoS, respectively. In general, the number

of mined or minted cryptocurrency decreases over time to reach zero; therefore, the global amount of the cryptocurrency is limited – this is a protection mechanism against inflation. The other source of the node's owner profit is the transaction fee collected from all transactions included in the newly added block.

The PoW principle is considered as the more secure and decentralised consensus mechanism. However, it is strongly eco-unfriendly and thus creates negative externalities. The energy consumption of competing for the reward nodes is enormous and negatively affects the environment even on the global scale. According to the Bitcoin Energy Consumption Index (digiconomist.net/bitcoin-energy-consumption/), the bitcoin network consumes 77.78 TWh electrical energy per year (estimation from February 2021). This is more than the energy consumption of some countries, e.g. Austria or Bangladesh and gives an incentive to intensive research for better consensus mechanisms, resulting in the number of their proposals.

The PoS principle does not cause any excessive energy consumption and can work faster than PoW. Nevertheless, it is criticised for worse decentralisation. As the probability of being awarded depends on the stake, thus “the rich get richer”. Their stakes can grow, which leads to the centralisation of the nodes that add transactions to the blockchain (Chitra and Evans, 2020). There are diverse solutions that alleviate the centralisation effect (He, Tand and Wang, 2020) applied in evolving blockchain networks. Moreover, PoS allows transaction throughput to be an order of magnitude higher compared to PoW. Consequently, PoS is more commonly used, and the best example is the already started transformation of the Ethereum network (ethereum.org/en/eth2/vision/) that moves from PoW to PoS. Ethereum is the second largest cryptocurrency by market cap. The investment cost in the node hardware in the case of PoS based network is significantly lower than in the case of PoW based network, where profit depends on hardware computation power. In PoS based network, the profit gets higher if the locked stake is bigger with respect to the stakes of other node owners (stakers). The staker is responsible for maintaining and securing the node. If the staker fails to do that or try to introduce a fraudulent transaction into blockchain, their entire stake might be at risk.

Consensus algorithms based on the Delegated-Proof-of-Stake (DPoS) principle become increasingly popular. The difference from PoS is that the stakers vote for the nodes (and their owners in that way) which can mint the new block. Next, the reward for the block is divided between the minter and the voters proportionally to their locked stake. There is a defined minimum stake value that allows to be the voter. The minting nodes belong to known and trusted parties. Their number is significantly lower than the number of staking nodes. In consequence, the blockchain can work faster and can process more transactions in a time. The responsibility for correct transaction processing rests with the minting nodes. The voter nodes still store all the blockchain and verify the correctness of newly added blocks. Each PoS or DPoS blockchain network, and any DeFi DApp that provides a staking reward, may apply a different algorithm of calculating staking rewards. The algorithm parameters can be adjusted dynamically, e.g. on a block-by-block basis. The parameters can include: the number of tokens the validator is staking, the time of staking, the age of tokens, the inflation rate of the currency, and others. All these make that the reward from staking varies a lot depending on the blockchain or DeFi DApp. Moreover, it varies over time, being influenced not only by changing number of the blockchain nodes and users but also by external factors. The enterprises that provide the technical infrastructure and maintenance for PoS nodes are called Staking Infrastructure Providers. Usually, they act as validators on several PoS blockchains, and they also provide Staking-as-a-Service for the clients that wish to delegate some funds to the provider's nodes (e.g. stakingteam.com, certus.one).

The technological setup described above gives space to enterprises that organise pools of mining or minting nodes. In the case of PoW blockchains, a concentration of computing hardware capacity increases the probability of higher income, which is later shared between the pool operator and node owners or lessees. The same in the case of PoS blockchains, a concentration of stakes belonging to a group of stakers increases the probability of getting a block award and gives more regular income. The operator can apply various technical solutions for the pool participation, even enabling staking by people without technical knowledge. However, a solo staking gives higher income compared to pool minting, where each successful block validation will split the rewards among the many participants of the pool and the operator. A staker can borrow money to make their stake bigger and earn more money – this is the basis for staking services, for technical and theoretical base compare (Park, Kim and Hong, 2020). The staker shares the profit with their lenders, creating in such a way the staking service. The enterprises that own some blockchain nodes and develop related software (e.g. cryptocurrency wallets) can easily offer staking services to the public.

We can distinguish two kinds of blockchain participants, those who run and maintain the network nodes and those who are just users that create transactions and possibly call smart contracts via crypto-wallet software. The first group bears the costs of the blockchain existence and security. The nodes can play different roles. In some networks, there are nodes that only store the ledger and nodes that also create new blocks. There can be nodes that validate groups of newly created blocks. In such a case, the profit from a new block is split between the creator and the validator that finalizes transactions. Ordinary users just pay for the network usage and usually do not get any income. However, in the case of some PoS and DPoS blockchains, it is possible to enable the users to stake a part of their tokens (to act as delegators) and participate in the gain from block creation and validation. Should a creator/validator misbehave, each of their delegators will be partially slashed in proportion to their delegated stake. This is why delegators should perform due diligence on delegates and spread their

stake over multiple delegates to mitigate the risks above. Smart contracts enable very flexible rule definitions for such participation, even including insurance against a risk related to the possible penalty for malfunctioning block processing.

Thanks to many cryptocurrency exchanges (see [Top Cryptocurrency Spot Exchanges coinmarketcap.com/rankings/exchanges/](https://coinmarketcap.com/rankings/exchanges/)), it is easy to change fiat money to a selected cryptocurrency or change the type of owned cryptocurrency. In consequence, it is easy to become a user of staking services. The exchanges own blockchain nodes for supported cryptocurrencies, and they have started to offer the staking services as well, proposing their customers to lock cryptocurrency on the exchange accounts. They even offer the staking of DeFi tokens. Depositing a currency on an exchange or other staking provider account is based on trust. The user must trust that the exchange won't get hacked, abscond with his assets, or incorrectly close out his position.

The PoS blockchains are complex and differ from each other. Any analysis of their long-term behaviour is not easy. There have been some attempts to build their models to get an analytical framework of them. Fanti, Kogan and Viswanath (2019) have developed a valuation framework for modelling a PoS financial system. Even though they applied several simplifications (e.g. they neglect the costs associated with block validation), they demonstrated an interesting approach to analysing the new complex systems. He, Tang and Wang (2020) have proposed another economic framework of PoS payment systems, and they analysed stake concentration in the blockchain networks. As a result, they provided some suggestions on improving the decentralisation level in a PoS blockchain network. They argue that when total network stake becomes significant, no individual stakeholder would have an incentive to stake individually because it is almost impossible to make a profit on its own. Consequently, they expect that pools of minting nodes will be dominant, as is the case of PoW blockchain networks.

Economic Perspective

As a result of the availability of blockchain and smart contract technology, several new financial instruments have been devised with the creation of cryptocurrencies, and existing financial assets have found their way into the world of cryptocurrencies. The low level of interest rates in developed economies in recent years results in a tendency to seek alternative ways of investing financial surpluses. It is most visible in the cryptocurrencies markets, however with the high level of volatility in terms of value investment, the risk is perceived to be high, so that the investors are advised to pay special attention to 'bad news' and moving patterns in order to make timely decisions on buying, selling or holding such assets (Luu Duc Huynh, 2019). It is also suggested that most crypto coins negatively change in terms of extreme value, thus making it difficult to manage portfolio in short time. Such a setup creates numerous opportunities for staking services that may prevent expected volatility or at least adjust the time framework of investment decisions and partially offset exchange risk. It may also improve the economic efficiency of cryptocurrency investment without harming the core incentives already applicable to crypto coins usage. Nevertheless, the perceived risk remains high, as it is associated both with the risk of underlying financial instrument (e.g. particular cryptocurrency), as well as intermediary risk typically involved. As staking services require reaching scale, it is usually provided by professional organisations that facilitate and actually offer these services to individuals. Organisational risk is therefore imminent, and questions related to reputation and trust emerge. One should also remember that this market is not only unregulated, but also, due to its complexity, exposed to several information asymmetries that increase final user risk exposition.

The market capitalisation of blockchain networks that allows for stacking is growing rapidly. While by mid-2020 it was estimated by European Blockchain Association (Schmidt, 2020) at 41.2 bln USD, according to data provided by www.stakingrewards.com by the end of February 2021, value of staked cryptocurrencies was estimated at the level of 99 bln USD, which amounts to some 6,3% of total value of cryptocurrencies in circulation, subject to their valuation in USD. The reward rate differs from almost zero up to 300% per annum, subject to the type of currency, platform provider, contract length and other conditions, with an average of 22%. The same service reports that the market cap of proof-of-stake blockchains grew to more than \$175 billion in 2020, while PoS blockchains were paid average annual returns of 11.2%. Obviously, such a high level of passive income does incorporate additional historical exchange rate gains, which have arisen from cryptocurrencies appreciation vis-a-vis traditional currencies. Additionally, it should be noticed that arrival of such a newly developed financial instrument is still an issue from the regulatory and especially taxation point of view, as staking actually combine numerous financial operations which have different fiscal status and effects (Landoni and Sutherland, 2020); and taxation practice varies from one country to another. Typically, services for individual users are offered by relatively young start-up companies, such as stakingfacilities.com, offering a comprehensive range of operations both individually tailored and related to b2b relations. As the service underlines, "participating in the maintenance of a PoS network is a complex task that requires a deep technological understanding of the underlying protocol, constant availability, industry-grade hardware and significant capital expenditure. Token holders who want to participate in the maintenance of the network and ultimately earn rewards & fees for doing so, but who are unwilling or unable to do so themselves, can use the services of a St-a-a-S (Staking as a service) by delegating their tokens. St-a-a-S provider charges a commission fee for their services."

Chitra and Evans (2020) have built a unifying model for studying the security risks of staking methods and concluded that the return profile of a portfolio of derivatives and staked assets resembles a portfolio of bonds and options on bonds. Thus, staking services can be compared to debt markets with their typical instruments, e.g. bonds. Extending such a framework on cryptocurrencies exchanges may allow comparing the second to traditional stock exchanges. Therefore, stocking can be viewed as a further step in fully emulating traditional investment markets in the future blockchain world. Moreover, as Roşu and Saleh (2021) claim, under certain assumptions, using PoS protocol, and thus stacking services, does not lead to excessive wealth accumulation and the rich getting richer but rather to stable investor shares.

Conclusions

The staking services are very new, and little research has been done on them. The crucial technical challenge is how to eliminate the risk of centralisation. Even though some mechanisms are already adopted in existing blockchain networks, their effectiveness is uncertain. The flexibility of smart contracts and the trust we can carry on them open wide space for shaping staking services, thus competitively developing them for very different end-user expectations. The problem is that many potential end-users do not understand programming languages; thus, a natural language description of smart contracts should be accompanied. Some research already provides an automatic translation of a smart contract into a natural language, but no satisfactory translator is yet available.

Today's reality of traditional deposit services with almost no interest, economic uncertainty of countries, and distrust of banks are factors that can ensure great incoming popularity of staking services. The services can take the form of Deposit-as-a-Service, and the mentioned factors make DaaS a promising business. It can gain popularity thanks to new specialized DeFi DApps, which are relatively easy to be developed. The further progress of stacking services in any form will lead to the emergence of a new competitive market. However, it is a big challenge to work out either legal regulations or just good market practices to protect the actors of the services, both the staking infrastructure providers and their customers.

Staking as a result of technological development sets a solid yet well-defined base for services similar to traditional deposits. Further, the development of their technological and financial aspects may bring additional spin-off services, analogically to various derivatives constructed in reference to underlying financial instruments, in this case, staking contracts. However, such development requires standardisation and well-defined transparent procedures with an adequate institutional framework. As described above, although the market seems to be well developed and matured from the technical point of view, numerous obstacles exist on both client and regulatory sides. Nevertheless, current involvement and on-hand experience with staking may prove to be a fruitful investment not only in terms of decent cryptocurrencies involved returns but also as a mean of wider financial market development.

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